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Revisiting Robustness in Graph Machine Learning [1]

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Graph Neural Networks (GNNs) suffer from [3] **over-robustness**

- Common threat models allow perturbations which change the node-level semantic content in graphs [4]
- GNNs are **wrongly robust** to these changes
- Introduce semantic-aware notions of robustness
- No robustness-accuracy tradeoff for inductively classifying a new node

On Adversarial Examples [5]

- Small perturbations (ℓ_p-norm) [6]
- Change prediction of classifier f
- Unchanged semantic content (e.g., image category)

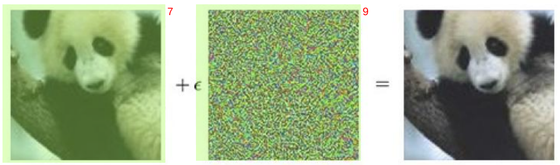
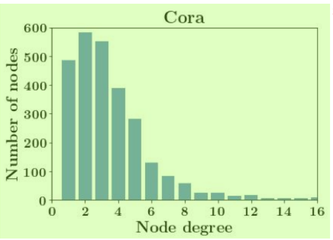


Figure: Goodfellow et al. (2015) [10]

What constitutes a small, semantics-preserving perturbation to a graph? [13]

Commonly: **Restrict** number of inserted / deleted edges (ℓ_0-norm) [14]



[15]

⚠ [16] Majority of nodes are low-degree [17]

→ ***Large effect*** on neighborhood [18]

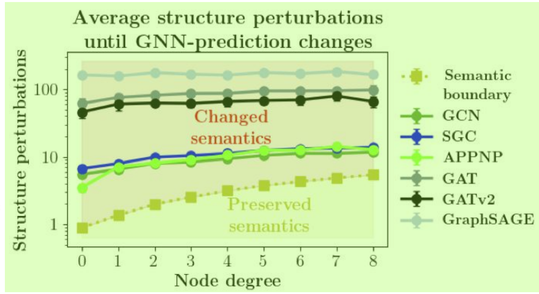
Do common threat models preserve semantic content? [19]

Use Random Graph Models [20]

- Exactly measure (node-level) [21] semantic change

New Phenomenon [22]

Over-robustness: Robustness *beyond* [23] semantic change



[24]

Semantic-Aware Robustness [25]

Idea: Use reference node classifier g to indicate semantic change. [26]

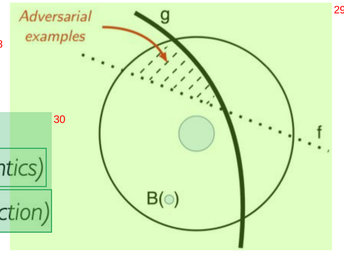
Adversarial Example [27]

A perturbed graph $\tilde{G} \in \mathcal{B}(G)$ is said to be *adversarial* for [28] a node classifier f at a target node v w.r.t. g iff:

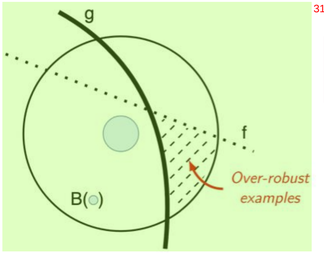
i. $f(G,y)_v = g(G,y)_v$ *(correct clean prediction)* [30]

ii. $g(\tilde{G},y)_v = g(G,y)_v$ *(perturbation preserves semantics)*

iii. $f(\tilde{G},y)_v \neq g(G,y)_v$. *(node classifier changes prediction)*



[29]



[31]

Over-robust Example [32]

A perturbed graph $\tilde{G} \in \mathcal{B}(G)$ is said to be an *over-robust example* for a node classifier f at a target node v w.r.t. g iff: [33]

i. $f(G,y)_v = g(G,y)_v$ *(correct clean prediction)* [34]

ii. $g(\tilde{G},y)_v \neq g(G,y)_v$ *(perturbation changes semantics)*

iii. $f(\tilde{G},y)_v = g(G,y)_v$. *(node classifier stays unchanged)*

Theorem (informal) [35]

There is no semantic-aware robustness-accuracy tradeoff for non-i.i.d. graph [36] data when classifying a newly added node to a graph.

Experiments [37]

Setup: Contextual stochastic block models (CSBM) [38]

- *Derived* Bayes optimal classifier as reference classifier g [39]

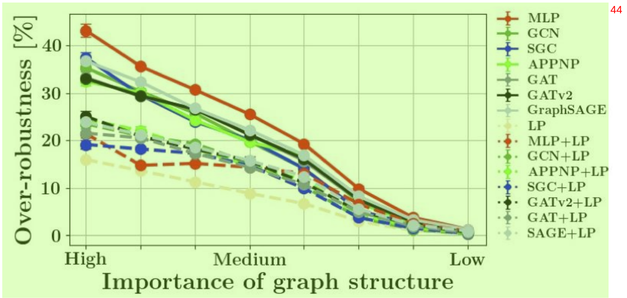
Table: Existence probability of graphs with changed semantics. [40]

Threat models	Importance of graph structure: High	·	·	Medium	·	·	Low	·
$\mathcal{B}_2(\cdot)$	36%	31%	26%	20%	14%	6%	2%	1%
$\mathcal{B}_{\text{deg}}(\cdot)$	**76%**	**60%**	**55%**	**49%**	40%	22%	9%	3%
$\mathcal{B}_{\text{deg+2}}(\cdot)$	100%	100%	99%	93%	81%	52%	25%	9%

[41]

→ **High prevalence** of graphs with changed semantics! [42]

How much of the measured robustness is in fact over-robustness? [43]



[44]

→ **Significant amount** of conventional robustness is **over-robustness** [45]

Take-Aways [46]

- Use strong attacks [47]
  → Reduce the amount of over-robustness measured
- Be cautious in applying ℓ_0-norm restricted adversaries
- **More research** into semantic-preserving perturbations **necessary**

[48]